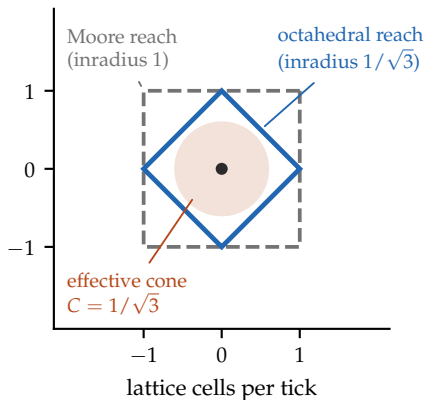
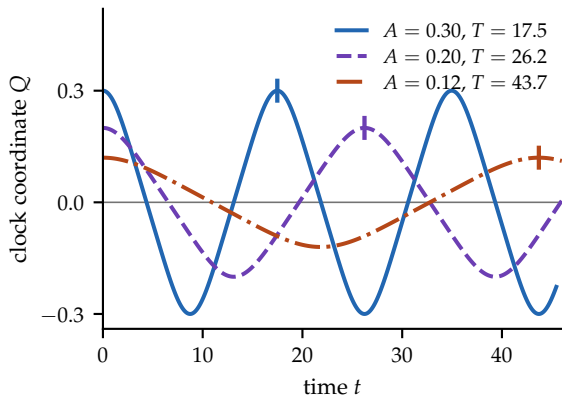


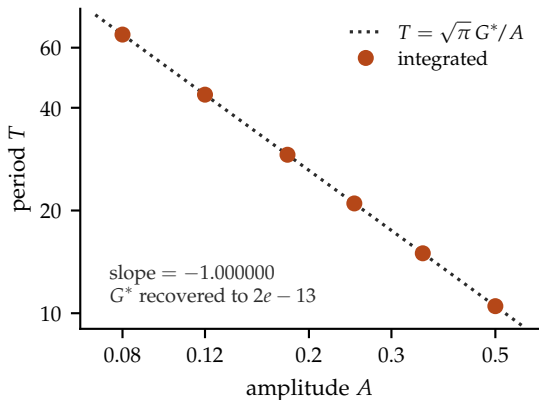
(a) substrate: the effective cone sits inside every candidate causal polytope



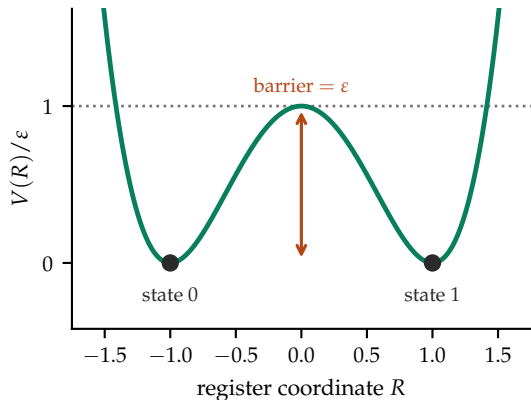
(b) clock: a bigger swing is a *faster* clock (ticks mark one period)



(c) ideal normal-form clock law, no fitted scale: slope -1 and the constant is G^*

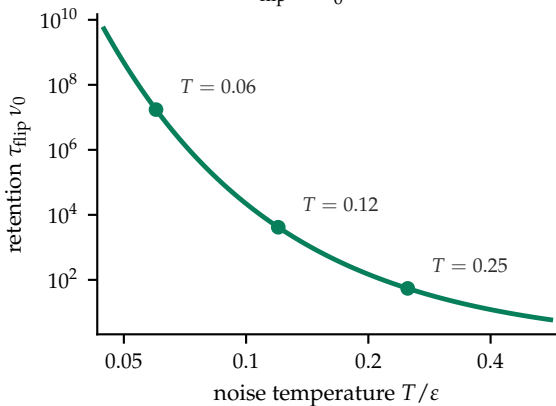


(d) register: two states, one barrier, and it is the same ϵ



(e) retention is Arrhenius:

$$\tau_{\text{flip}} \sim \nu_0^{-1} e^{\epsilon/T}$$



(f) gate: the classical threshold statistic interpolates with the timescale ratio

